

Claire Jing

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EDUCATION

Georgia Institute of Technology

Atlanta, GA

Bachelor of Science in Mathematics and Computing

May 2028

Courses: Probability, Statistics, Linear Algebra, ODE, Object-Oriented Programming

GPA: 4.0/4.0

EXPERIENCE

Software Engineering Intern

Georgia Tech Research Institute

June 2026 – Present

- Developed a mathematical formalization pipeline to ingest LaTeX theses & extract structured content via BAML.
- Engineered a graph isomorphism tool for directed acyclic graphs (DAGs) using Python and NetworkX, implementing configurable metadata-preserving matching rules to compare graph structures.
- Optimized auto-formalization workflow by 40% by designing a BAML-backed extraction layer that maps mathematical dependencies to formal code.

Quantitative Analyst

GT Quant Sector

April 2026 – Present

- Manage quant sleeve of a \$2.7M endowment, developing models for alpha generation and portfolio construction.
- Develop a state-space model with Kalman filtering to dynamically estimate market and sector betas and generate mean-reversion signals from return prediction errors.
- Build data, backtesting, and parameter-optimization pipelines incorporating EWMA z-scores, AR(1) dynamics, and rolling autocorrelation across individual equities and sectors.

Undergraduate Researcher

Georgia Institute of Technology

Jan. 2026 – May 2026

- Modeled dynamical systems for VR/AR using SwiftUI to create interactive educational physics simulations.
- Built COMSOL bridge that translates hardware data into VR experience, streamlining data conversion by 25%.
- Developed sensory mapping engine with SwiftUI that accelerates visualization process on Apple Vision Pro by 35%.

General Analyst

Undergraduate Consulting Club

Jan. 2026 – April 2026

- Built an ETL pipeline to standardize fiscal data for startup company Ignea, reducing reporting latency by 30%.
- Synthesized financial streams into centralized Tableau dashboards tracking expansion KPIs & overhead costs.

PROJECTS

Market Phase Transitions | *Python, NumPy, Pandas, Matplotlib, SciPy*

- Developed a Python-based market regime detection model using Random Matrix Theory, identifying 5 significant factors across 20+ years of S&P 500 equity and Treasury data.
- Engineered a rolling spectral-gap indicator to quantify market-wide synchronization, capturing sharp spikes during the 2008 financial crisis and COVID-19 shock.
- Reduced 10 Treasury maturities into 3 principal yield-curve drivers, explaining 92.1% of total market movement.

Gen Z Market Resonance | *Python, SQL, NumPy, Statsmodels*

- Analyzed trading behavior across 10,000+ Gen Z-coded accounts over 90+ days alongside Dr. Watkins @ UGA.
- Increased explanatory power of volatility-correlation models ($\Delta R^2 + 0.05$) with a sentiment-based signal processing tool to extract behavioral signals and identify decision-making patterns correlated with market volatility events.

Metro Affordability | *SQL, Python, AWS, Tableau*

- Identified neighborhoods in NYC with the steepest post-pandemic rent growth by isolating 3 ZIP codes with 20%+ YoY increases, analyzing years of housing-market data using Python and SQL-backed repeated queries.
- Automated data pipelines using SQL for dashboards, leveraging AWS to host datasets and track regional metrics.

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, SQL, Swift, Lean, C++

Frameworks & Libraries: React, Node.js, Spring Boot, Spring MVC, JPA, NumPy, Pandas, SciPy, Matplotlib

Developer Tools: Git, GitHub, Bash, Linux/Unix, Jupyter, VS Code, AWS (S3, QuickSight), Tableau, Excel

HONORS AND DISTINCTIONS

Center for Finance & Technology Scholar, *Georgia Institute of Technology*

Jane Street Young Women in Mathematics Winner, *Mathematical Association of America*

2x American Invitational Mathematics Examination (AIME) Qualifier, *Mathematical Association of America*